# Software Requirements Specification (SRS)

Asset & Resource Allocation Management System (ResourceX)

## Preface

This document defines the software requirements for the Asset & Resource Allocation Management System (ResourceX). It is intended for students, instructors, project reviewers, developers, and testers.

### Version History

| Version | Date | Description |
|---|---|---|
| 1.0 | Initial | First version of the Software Requirements Specification created for academic submission. |
| 2.0 | 21 October 2026 | Updated to match the implemented system: the single Staff actor is split into Inventory, Maintenance and Procurement Officers; Platform Owner and Platform Employee actors added for the multi-tenant SaaS platform; organization registration, subscriptions, support tickets, announcements, notifications, return deadlines and extensions, role-coded email formats and email/password login added; use case diagram redrawn. |

Rationale: This document has been created as part of the Requirement Engineering process to formally specify system requirements.

## 1. Introduction

### 1.1 Need for the System

Organizations often rely on manual or fragmented methods to manage assets and resources. This results in inefficient utilization, lack of transparency, delayed approvals, and difficulty in

tracking maintenance and allocation history.

## 1.2 System Overview

The Asset & Resource Allocation Management System is a centralized web-based application that manages asset tracking, allocation requests, approval workflows, maintenance, returns, scrap and procurement activities.

The system is a **multi-tenant SaaS platform**. Many organizations (institutions) are onboarded onto one platform, and each organization's data is isolated from the others. The platform itself is run by a Platform Owner and Platform Employees, who onboard, support and bill the organizations.

## 1.3 Actors

The system supports nine actors. Actors 1–7 work inside a single organization. Actors 8–9 work at the platform level, across organizations.

| # | Actor | Level | Responsibilities |
|---|---|---|---|
| 1 | Requestor | Organization | Requests resources, tracks requests, uses allocated resources, returns them, requests maintenance and return-deadline extensions. |
| 2 | Department Head | Organization (one per department) | Approves or rejects requests and extension requests, monitors department resources and stock, raises procurement requests. |
| 3 | Registrar | Organization (no department) | Approves or rejects procurement requests and views organization-wide analytics. |
| 4 | Inventory Officer | Organization (department) | Allocates resources, registers new resources, tracks return deadlines, inspects returned resources and routes them to stock, maintenance or scrap. |
| 5 | Maintenance Officer | Organization (department) | Handles maintenance tickets, marks resources repaired or scrapped, keeps maintenance history. |
| 6 | Procurement Officer | Organization (department) | Processes approved procurement requests, creates purchase orders, records vendor, invoice and warranty details, tracks delivery. |
| 7 | System Admin | Organization | Manages users, departments, subscription plan, support tickets and announcements for their own organization. |
| 8 | Platform Employee | Platform | Supports the organizations assigned to them: resolves support tickets, manages organization status, sends announcements. |

| | | | |
|---|---|---|---|
| 9 | Platform Owner | Platform | Owns the platform: approves organization registrations, manages employees, organizations, subscription plans and revenue. |

*The Inventory, Maintenance and Procurement Officers replace the single "Staff" actor used in version 1.0. Requestors and Registrars are not tied to a department. Platform Owner and Platform Employee are not tied to a department or organization.*

## 1.4 System Functions (Brief)

The system provides the following core functions.

**F1 Resource Request Management** – Requestors submit resource requests, track their status and view request history.

**F2 Request Approval Management** – Department Heads review incoming requests and approve or reject them.

**F3 Resource Allocation Management** – Inventory Officers allocate resources to approved requests, set return deadlines and notify requestors.

**F4 Resource Inventory Management** – A central inventory of all resources per organization; Inventory Officers register and update resources.

**F5 Procurement Management** – Department Heads raise procurement requests, Registrars approve them and Procurement Officers complete the purchase.

**F6 Maintenance Management** – Requestors report issues and Maintenance Officers repair or scrap resources.

**F7 Resource Return Management** – Requestors return resources; Inventory Officers inspect them and route them to stock, maintenance or scrap.

**F8 Return Deadlines and Extensions** – Allocated resources carry a return deadline; requestors can ask for an extension, which the Department Head approves or rejects.

**F9 Notifications** – Users are notified of workflow events, approaching or missed deadlines, and low stock.

**F10 Analytics and Monitoring** – Dashboards for department, organization and platform-level statistics.

**F11 User Management** – System Admins create and manage the users of their organization.

**F12 Department Management** – System Admins create and manage departments.

**F13 Organization Registration and Management** – Organizations register on the platform; the Platform Owner approves them and assigns a Platform Employee.

**F14 Subscription and Revenue Management** – Organizations subscribe to a plan; the Platform Owner tracks invoices and revenue.

**F15 Support and Communication** – Organizations raise support tickets; Platform Employees resolve them and send announcements.

**F16 Platform Employee Management** – The Platform Owner creates and manages Platform Employee accounts.

### 1.5 Interaction with Other Systems

The system is a standalone web application with a React front end and a NestJS REST API documented with Swagger (OpenAPI). It may integrate with external authentication, payment or inventory systems in the future.

### 1.6 Business and Strategic Objectives

- Improve asset utilization
- Increase transparency and accountability
- Reduce operational delays
- Support informed decision-making
- Offer the system as a subscription service to many organizations

## 2. Glossary

| Term | Definition |
|---|---|
| Asset / Resource | A physical resource such as a laptop, projector, router, or other equipment. |
| Allocation | Assignment of a resource to a requestor against an approved request. |
| Registrar | Senior authority in an organization responsible for procurement approvals. |
| Organization / Institution (Tenant) | A customer of the platform whose users and data are isolated from other organizations. |
| Platform | The ResourceX service that hosts all organizations, run by the Platform Owner and Platform Employees. |
| Subscription Plan | A paid plan (Starter, Professional, Enterprise) that sets the user limit and features of an organization. |
| Indent / Procurement Request | A request from a Department Head to purchase additional resources. |
| Purchase Order (PO) | An order placed with a vendor by the Procurement Officer after a procurement request is approved. |
| Threshold | The minimum stock level for a resource type; stock below it is Low Stock. |

| | |
|---|---|
| Return Deadline | The date by which an allocated resource must be returned. |
| Support Ticket | An issue raised by an organization for the platform team to resolve. |
| Announcement | A message sent by the platform team to one or more organizations. |
| Audit Log | A record of system activities for accountability. |

# 3. User Requirements Definition

This section describes the services provided to system users. Requirements are expressed in natural language and are understandable to non-technical stakeholders.

## 3.1 Functional Requirements

The functional requirements describe the operations and services that the system must perform.

### 3.1.1 Authentication Module

**FR-1** The system shall let users sign in with email and password only; the dashboard is chosen from the account's role, with no role selection.

**FR-2** The system shall restrict every operation to the roles allowed to perform it (role-based access control).

**FR-3** The system shall restrict organization-level users to the data of their own organization.

### 3.1.2 Resource Request Module

**FR-4** The system shall allow requestors to submit resource requests including department, resource type, quantity, and reason.

**FR-5** The system shall generate a unique request ID for each request.

**FR-6** The system shall allow requestors to request a new resource type if the resource does not exist.

**FR-7** The system shall forward submitted requests to the Department Head of the selected department.

**FR-8** The system shall allow requestors to view all their submitted requests and search them by request ID.

**FR-9** The system shall display request status: Pending, Approved, Rejected, Allocated, and Completed.

**FR-10** The system shall allow requestors to confirm receipt of an allocated resource.

**FR-11** The system shall maintain history of completed and rejected requests.

### 3.1.3 Request Approval Module

**FR-12** The system shall allow Department Heads to view incoming requests for their department.

**FR-13** The system shall allow Department Heads to approve requests, forwarding them to the Inventory Officer for allocation.

**FR-14** The system shall allow Department Heads to reject requests and notify the requestor.

#### 3.1.4 Resource Allocation Module

**FR-15** The system shall allow Inventory Officers to view approved requests awaiting allocation.

**FR-16** The system shall allow Inventory Officers to assign one or more available resources to an approved request.

**FR-17** The system shall record allocation details including request ID, resource code, and return deadline.

**FR-18** The system shall notify the requestor and Department Head when resources are allocated.

**FR-19** The system shall maintain allocation history records.

#### 3.1.5 Resource Inventory Module

**FR-20** The system shall allow Inventory Officers and Department Heads to view the resource inventory, searchable by serial number or resource code.

**FR-21** The system shall allow Inventory Officers to register new resources, including newly purchased ones.

**FR-22** The system shall allow Inventory Officers to update resource information.

**FR-23** The system shall track resource status: Available, Allocated, Maintenance Requested, Maintenance, Repaired, Returned, and Scrapped.

**FR-24** The system shall maintain resource lifecycle history.

**FR-25** The system shall allow Department Heads to set stock thresholds and show stock as Safe, Near Threshold, or Low, searchable by type and filterable by status.

#### 3.1.6 Return Deadline and Extension Module

**FR-26** The system shall show return deadlines for all allocated resources and highlight overdue resources.

**FR-27** The system shall allow Inventory Officers to force the return of overdue resources.

**FR-28** The system shall allow requestors to request an extension of a return deadline.

**FR-29** The system shall allow Department Heads to approve or reject extension requests.

#### 3.1.7 Procurement Management Module

**FR-30** The system shall allow Department Heads to create procurement requests stating resource type and quantity.

**FR-31** The system shall allow the Registrar to view, approve, and reject procurement requests.

**FR-32** The system shall make approved procurement requests available to the Procurement Officer of that department.

**FR-33** The system shall allow Procurement Officers to create purchase orders with vendor, purchase date, invoice number, unit price, and warranty.

**FR-34** The system shall track procurement status: Pending Approval, Approved, PO Created, Out for Delivery, Fulfilled, Registered, and Rejected.

**FR-35** The system shall allow received resources to be registered into inventory.

**FR-36** The system shall maintain procurement and purchase history.

#### 3.1.8 Maintenance Management Module

**FR-37** The system shall allow requestors to submit maintenance requests for allocated resources.

**FR-38** The system shall allow Maintenance Officers to review maintenance requests, including those routed from returns, and set priority.

**FR-39** The system shall allow Maintenance Officers to mark resources as repaired, recording resolution notes, condition, and cost.

**FR-40** The system shall allow Maintenance Officers to mark resources as scrap if not repairable.

**FR-41** The system shall notify requestors about maintenance outcomes.

**FR-42** The system shall maintain maintenance history, filterable by outcome and searchable.

#### 3.1.9 Resource Return Module

**FR-43** The system shall allow requestors to submit return requests for allocated resources.

**FR-44** The system shall allow Inventory Officers to inspect returned resources and grade their condition.

**FR-45** The system shall allow Inventory Officers to route a returned resource to Available stock, the maintenance queue, or Scrap.

**FR-46** The system shall maintain return history records, visible to Inventory Officers and to the requestor who returned the resource.

#### 3.1.10 Notifications Module

**FR-47** The system shall notify the relevant users of request, approval, allocation, procurement, return, and maintenance events.

**FR-48** The system shall notify users of approaching and overdue return deadlines and procurement target dates.

**FR-49** The system shall send the Department Head one alert when a resource type falls below its threshold.

**FR-50** The system shall show an unread count and allow users to mark notifications as read.

#### 3.1.11 Analytics and Monitoring Module

**FR-51** The system shall allow Department Heads to view department resource analytics.

**FR-52** The system shall allow the Registrar to view organization-wide analytics.

**FR-53** The system shall allow the Platform Owner to view platform analytics: organizations, active users, subscriptions, and revenue.

#### 3.1.12 User Management Module

**FR-54** The system shall allow System Admins to create, edit, and deactivate users in their organization.

**FR-55** The system shall reject an email address that is already in use.

**FR-56** The system shall require organization-level emails in the format {name}{roleCode}{organizationName}@gmail.com, where the role code is re, rg, sa, dh, mo, io, or po.

**FR-57** The system shall not assign a department to Requestors or Registrars.

**FR-58** The system shall allow only one Department Head per department.

**FR-59** The system shall allow System Admins to search users by name or email and filter by role and status.

#### 3.1.13 Department Management Module

**FR-60** The system shall allow System Admins to create departments by name and generate a unique department ID per organization.

**FR-61** The system shall allow System Admins to edit and delete departments.

**FR-62** The system shall assign the department head when a Department Head user is created for that department.

#### 3.1.14 Organization Registration Module

**FR-63** The system shall allow a new organization to register with its name, a proof document for the organization, the System Admin's name, email, proof document and password, and a chosen subscription plan.

**FR-64** The system shall accept proof documents as PDF, PNG, or JPG up to 5 MB.

**FR-65** The system shall allow the Platform Owner to review registrations, download the proofs, and approve them by assigning a Platform Employee.

#### 3.1.15 Organization Management Module

**FR-66** The system shall allow the Platform Owner and Platform Employees to view and update organization details.

**FR-67** The system shall allow organizations to be activated, suspended, or reactivated.

#### 3.1.16 Subscription and Revenue Module

**FR-68** The system shall offer subscription plans with a user limit, yearly price, and feature list.

**FR-69** The system shall allow System Admins to view their current plan and change it.

**FR-70** The system shall allow the Platform Owner to view invoices, renewals, expirations, and revenue.

#### 3.1.17 Support and Communication Module

**FR-71** The system shall allow System Admins to raise support tickets for their organization.

**FR-72** The system shall allow Platform Employees to resolve tickets for their assigned organizations and update their status.

**FR-73** The system shall allow Platform Employees and the Platform Owner to send announcements to one organization or to all organizations.

**FR-74** The system shall allow System Admins to view announcements sent to their organization.

#### 3.1.18 Platform Employee Management Module

**FR-75** The system shall allow the Platform Owner to create Platform Employee accounts with emails of the form employee{number}@resourcex.com.

**FR-76** The system shall allow the Platform Owner to assign organizations to employees and to activate, suspend, or remove employee accounts.

#### 3.1.19 System Logs Module

**FR-77** The system shall record system activities such as requests, allocations, maintenance actions, and user updates.

**FR-78** The system shall log API requests and errors per module for auditing.

### 3.2 Non-Functional Requirements

The non-functional requirements describe the quality attributes of the system.

**3.2.1 Performance** – The system should respond to user actions within two seconds.

**3.2.2 Security** – The system shall use token-based (JWT) authentication and role-based access control, and shall validate all input.

**3.2.3 Multi-tenancy** – The system shall keep each organization's data isolated. Platform-level actors get controlled cross-organization access for onboarding, support, and billing only.

**3.2.4 Scalability** – The system should support increasing numbers of organizations, users, and resources without performance degradation.

**3.2.5 Reliability** – The system should maintain accurate resource records and prevent duplicate entries such as duplicate emails.

**3.2.6 Usability** – The system should provide a clear, role-specific dashboard for each actor.

**3.2.7 Maintainability** – The system should use a modular architecture, with one backend module per feature area.

**3.2.8 Availability** – The system should ensure high availability with minimal downtime.

**3.2.9 Compatibility** – The system should support modern web browsers and responsive layouts.

## 4. System Architecture

The system follows a web-based client–server architecture:

**Presentation Layer:** A React single-page application with a separate dashboard for each of the nine actors.

**Application Layer:** A NestJS REST API containing business logic, workflows, authentication (JWT), role guards, and per-module validation, logging, and security middleware. The API is documented with Swagger (OpenAPI 3.0).

**Data Layer:** Storage for organizations, subscription plans, users, departments, resources, requests, procurements, maintenance records, returns, notifications, support tickets, announcements, and audit logs. Every organization-level record carries an organization ID.

Reusable components include authentication, role-based authorization, and tenant scoping.

## 5. System Requirements Specification

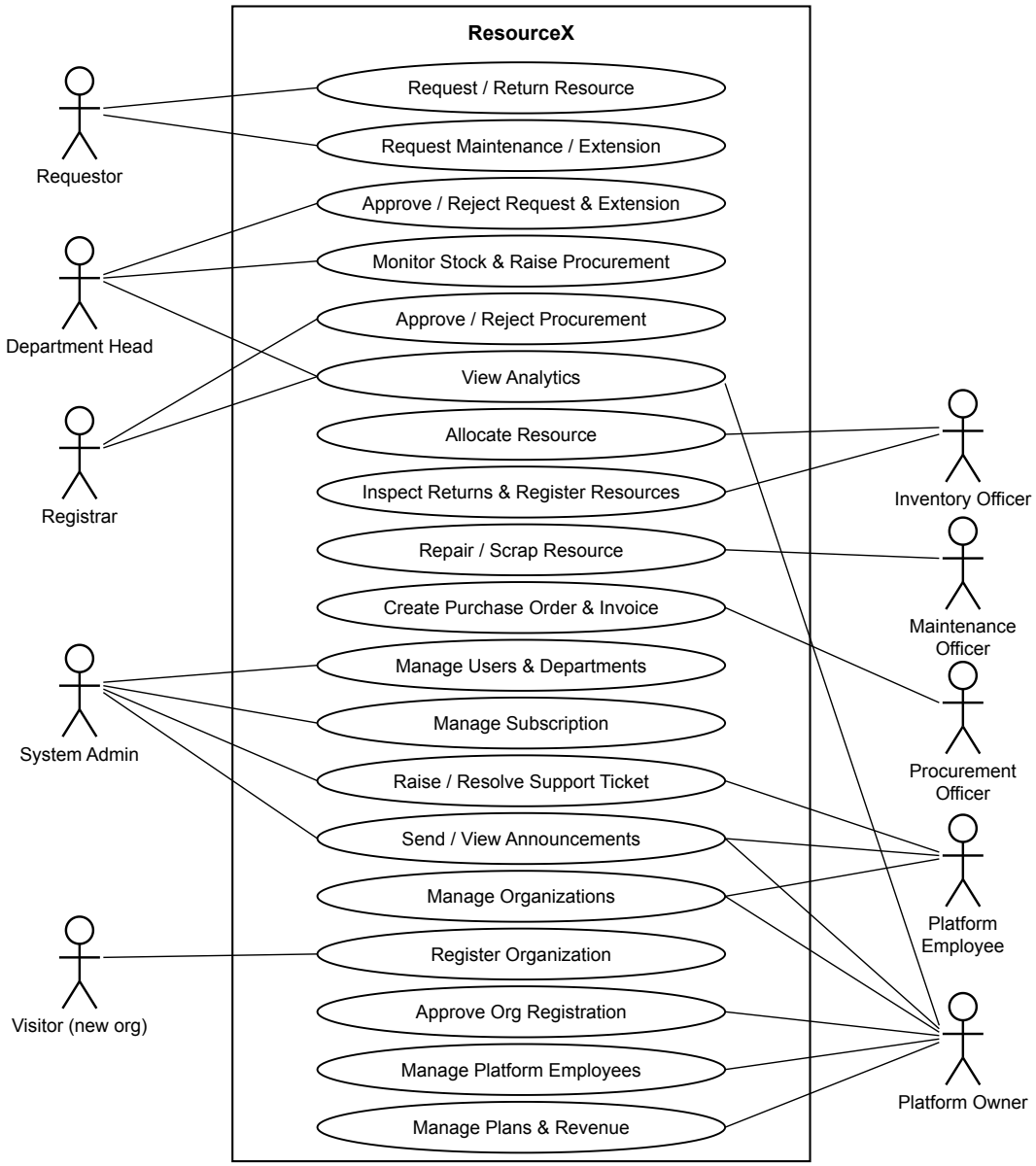
The functional and non-functional requirements in Section 3 form the system requirements. The REST endpoints that implement them are listed in the project's Swagger documentation.

## 6. System Models

The following system models are used.

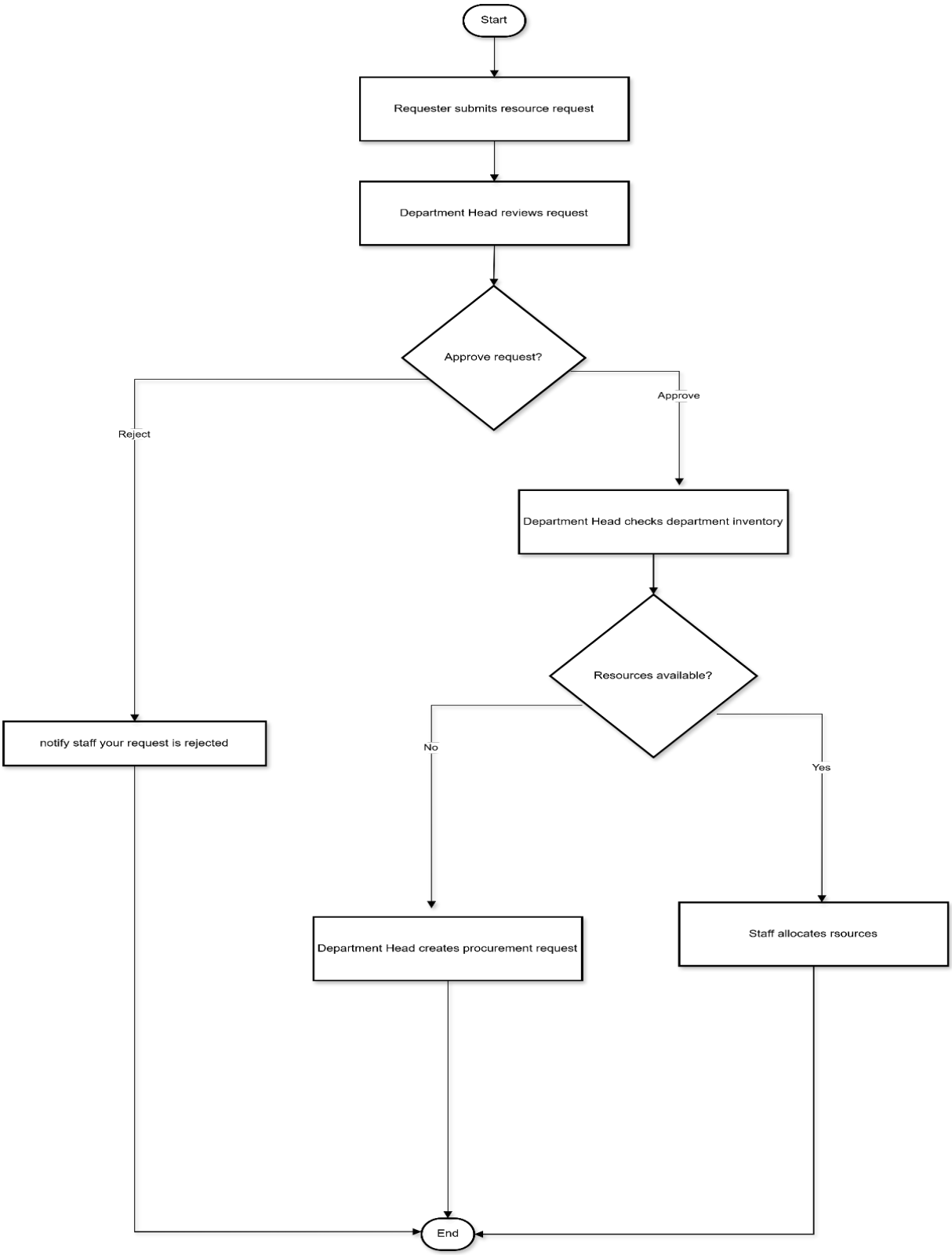
*The use case diagram below reflects version 2.0. The activity, sequence, and ER diagrams that follow are from version 1.0 and describe the core organization workflow. In them, "Staff" stands for the Inventory, Maintenance, or Procurement Officer, whichever owns that step. The data added for the platform level is listed in a table after the ER diagram.*

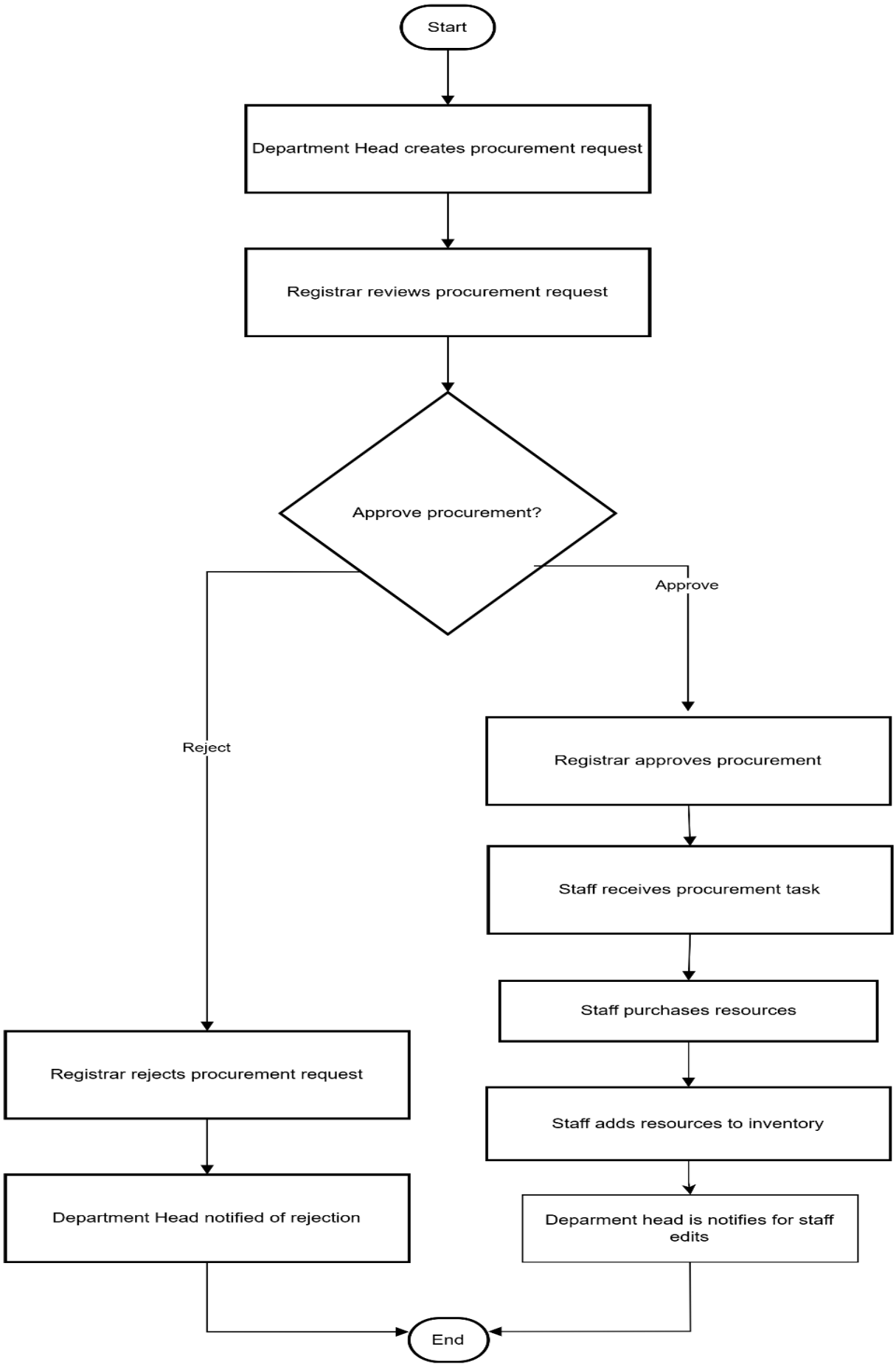
## Use Case Diagram

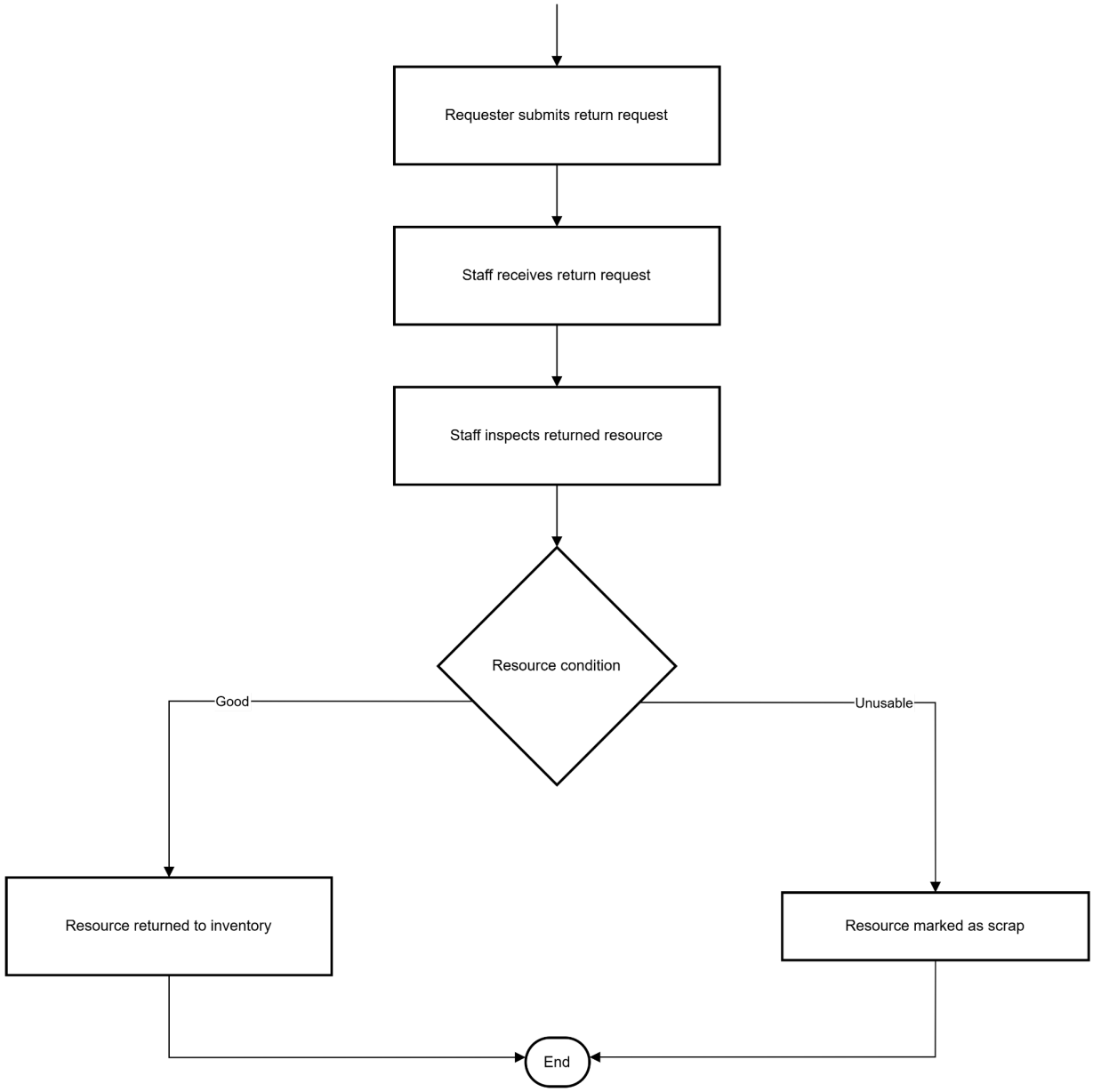



*"Visitor (new org)" is a person registering a new organization before it has any accounts; after approval they sign in as that organization's System Admin.*

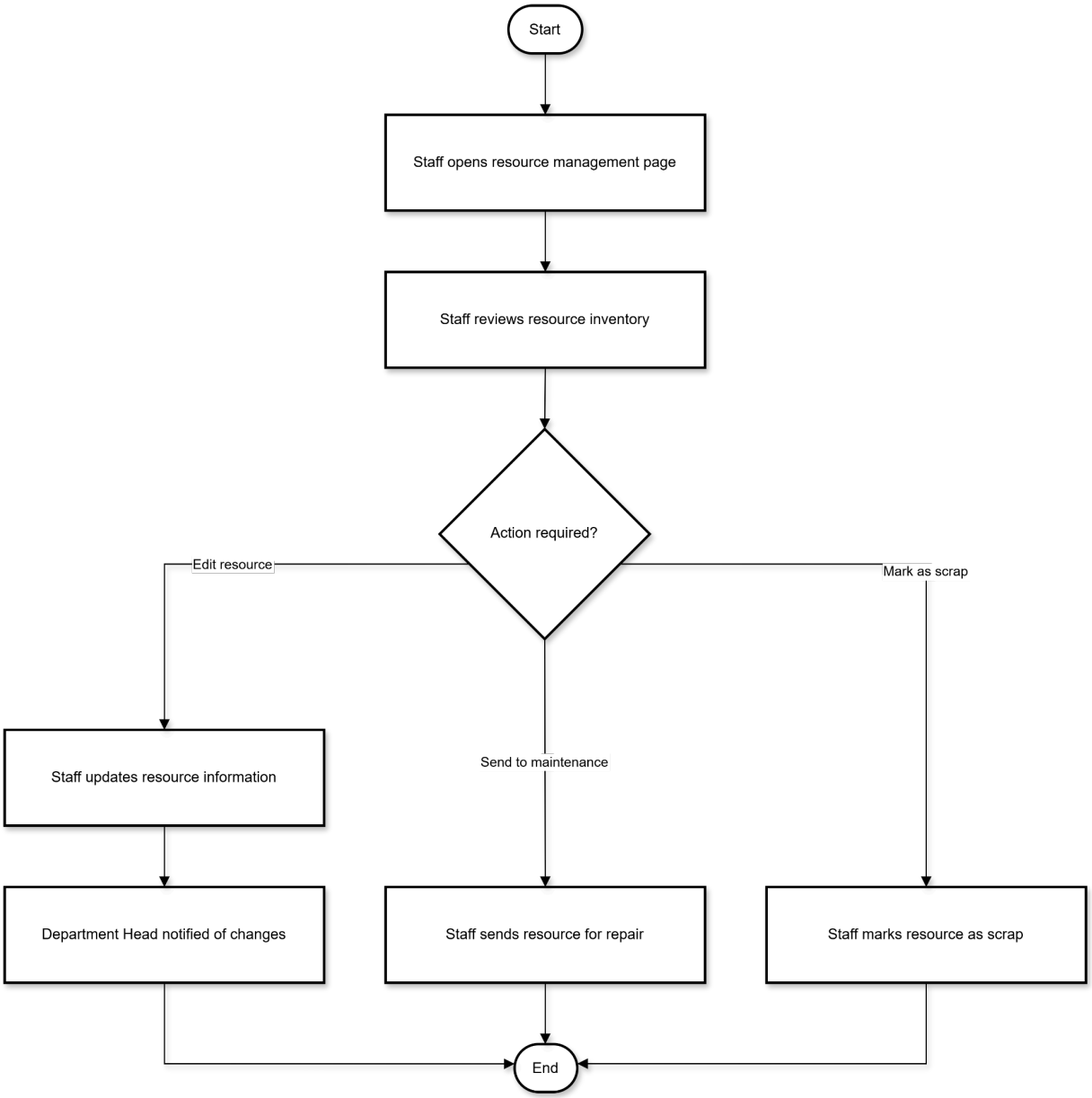
## Activity Diagrams


Start
Requester submits resource request
Department Head reviews request
Approve request?
Reject
Approve
Department Head checks department inventory
Resources available?
notify staff your request is rejected
No
Yes
Department Head creates procurement request
Staff allocates rsources
End

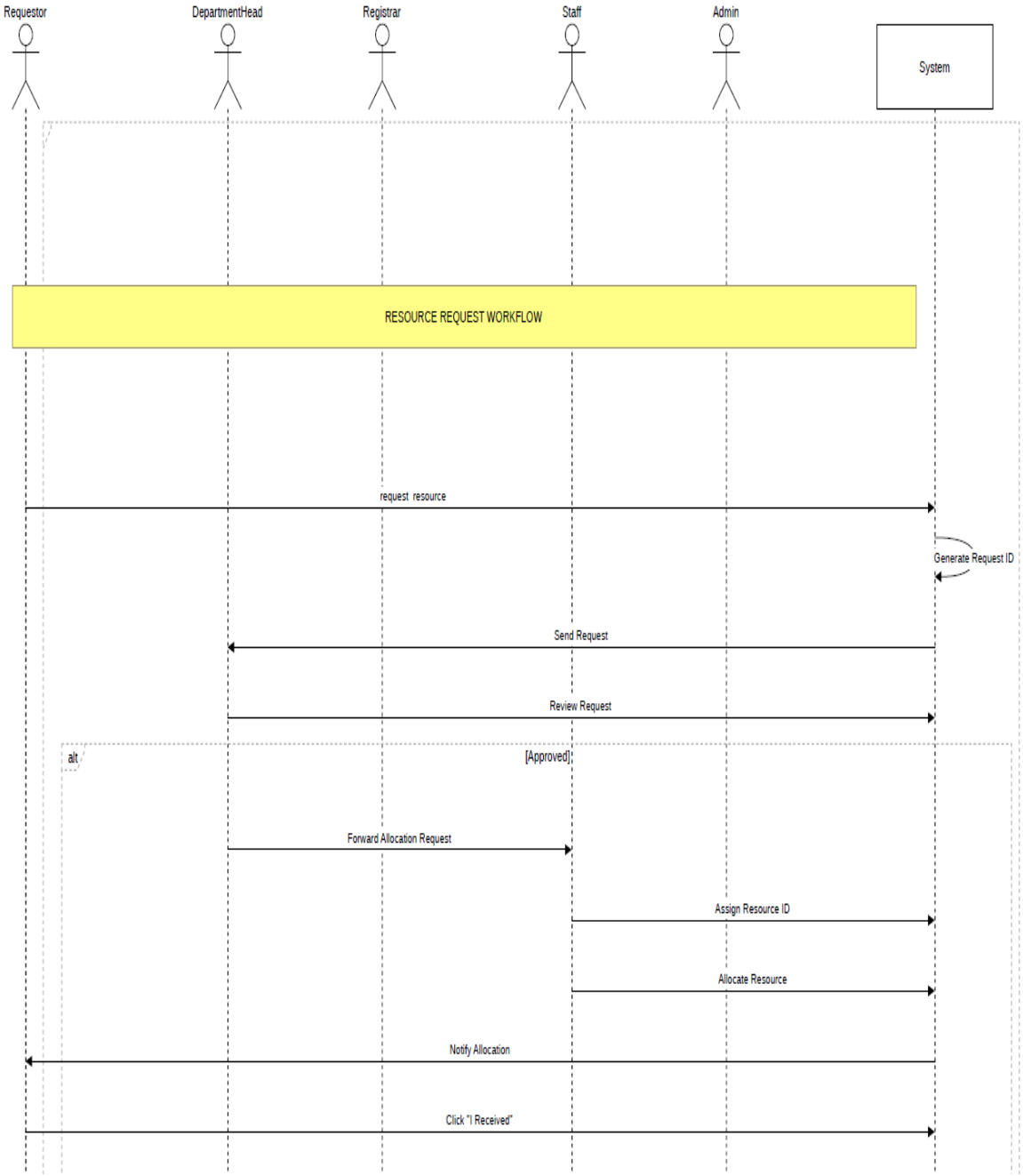

Start
Department Head creates procurement request
Registrar reviews procurement request
Approve procurement?
Approve
Reject
Registrar approves procurement
Staff receives procurement task
Staff purchases resources
Staff adds resources to inventory
Deparment head is notifies for staff edits
Registrar rejects procurement request
Department Head notified of rejection
End

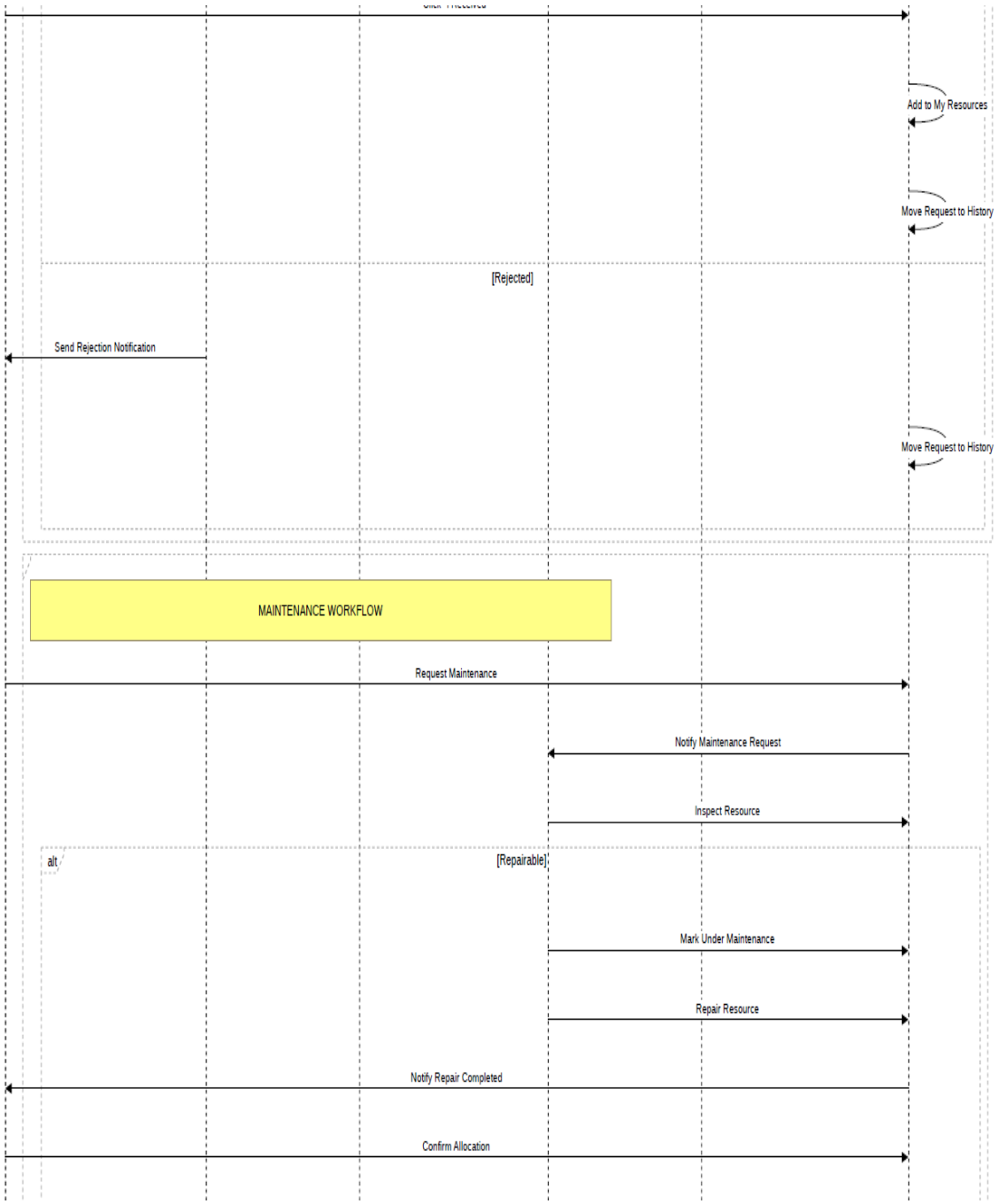

Requester submits return request
Staff receives return request
Staff inspects returned resource
Resource condition
Good
Unusable
Resource returned to inventory
Resource marked as scrap
End

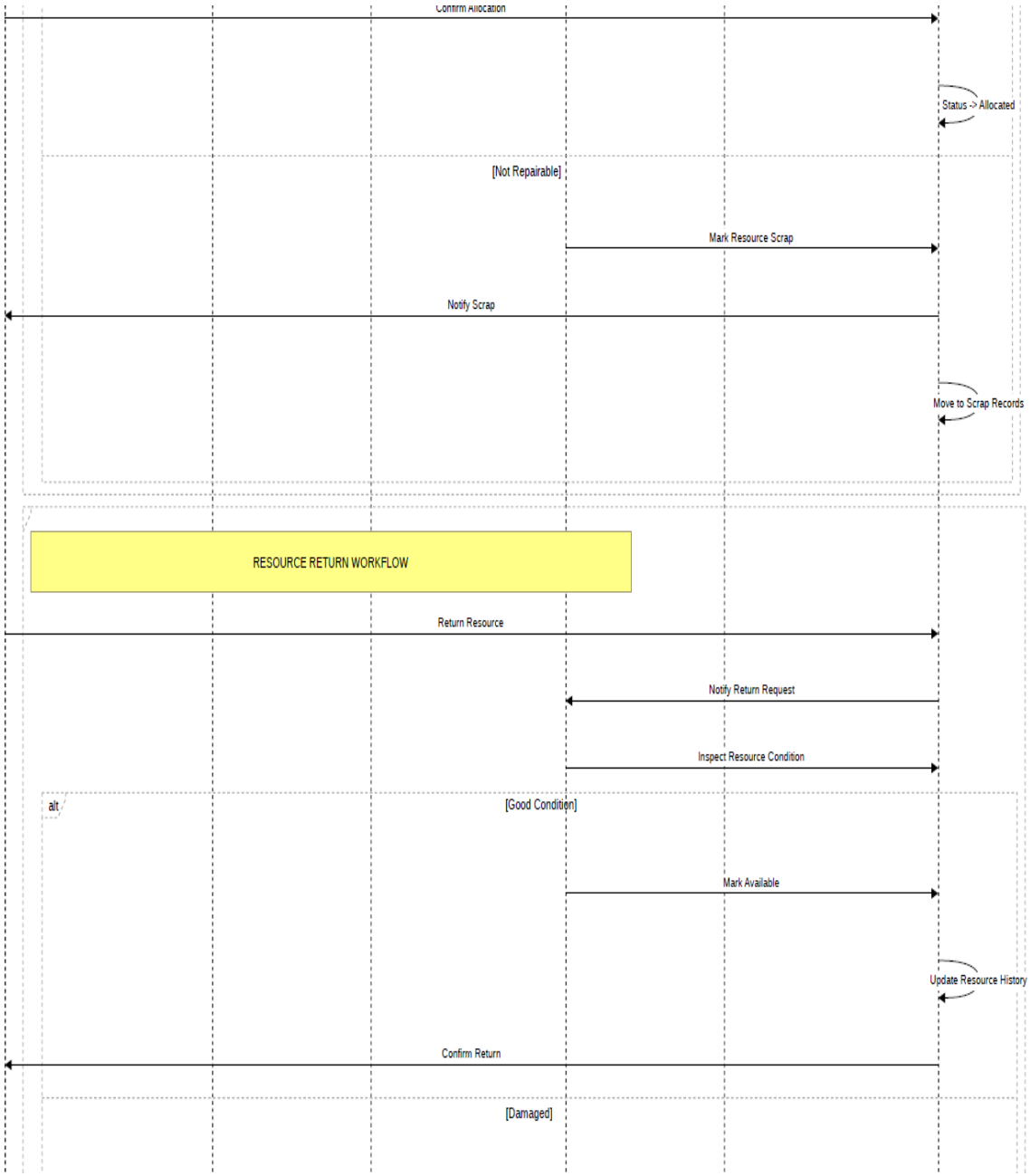

Start
Requester reports resource issue
Staff receives maintenance request
Staff inspects resource
Repair possible?
Yes
No
Staff repairs resource
Resource marked as scrap
Requester notified resource repaired
Requester notified resource scrapped
End

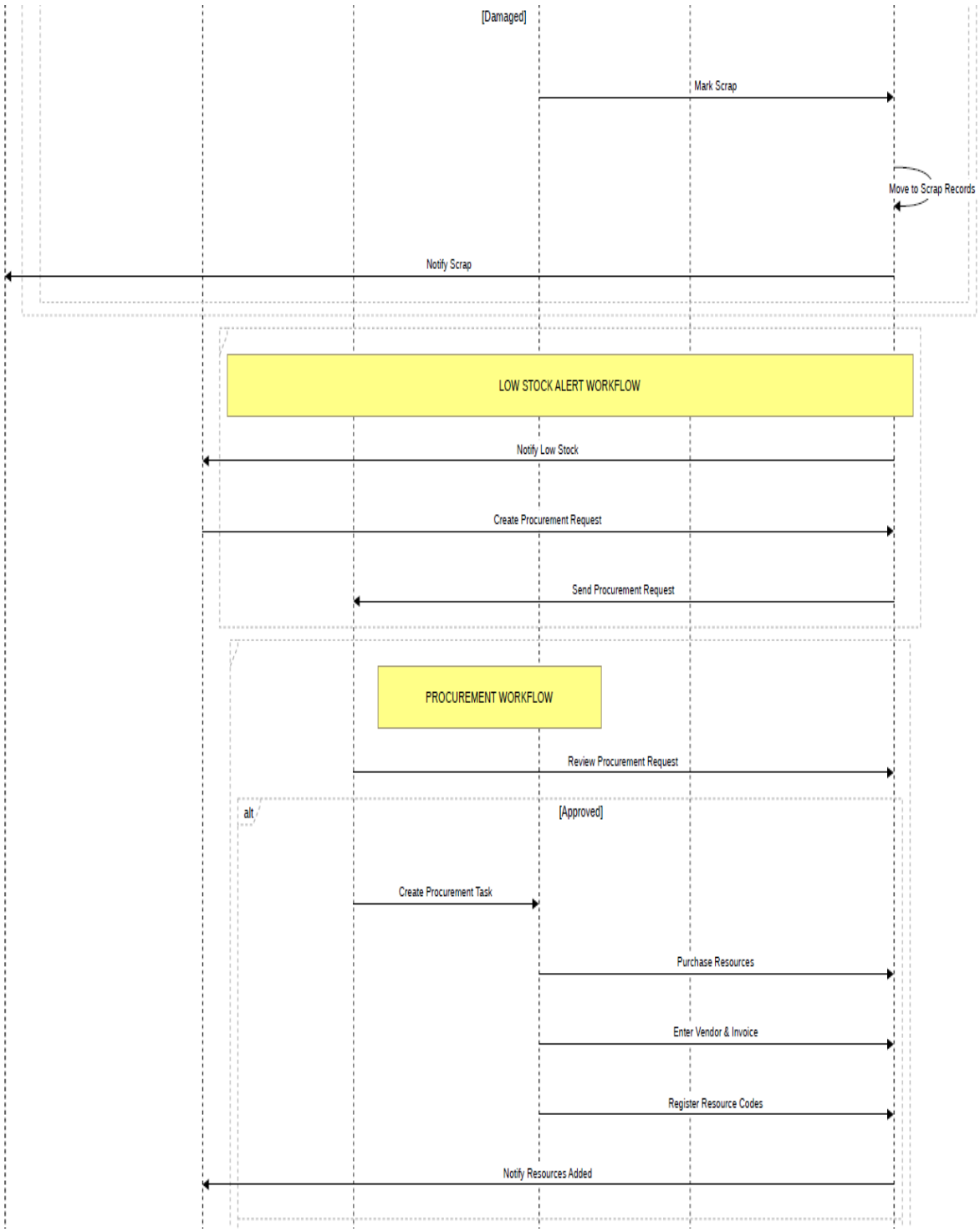

Start
Staff opens resource management page
Staff reviews resource inventory
Action required?
Edit resource
Mark as scrap
Send to maintenance
Staff updates resource information
Department Head notified of changes
Staff sends resource for repair
Staff marks resource as scrap
End

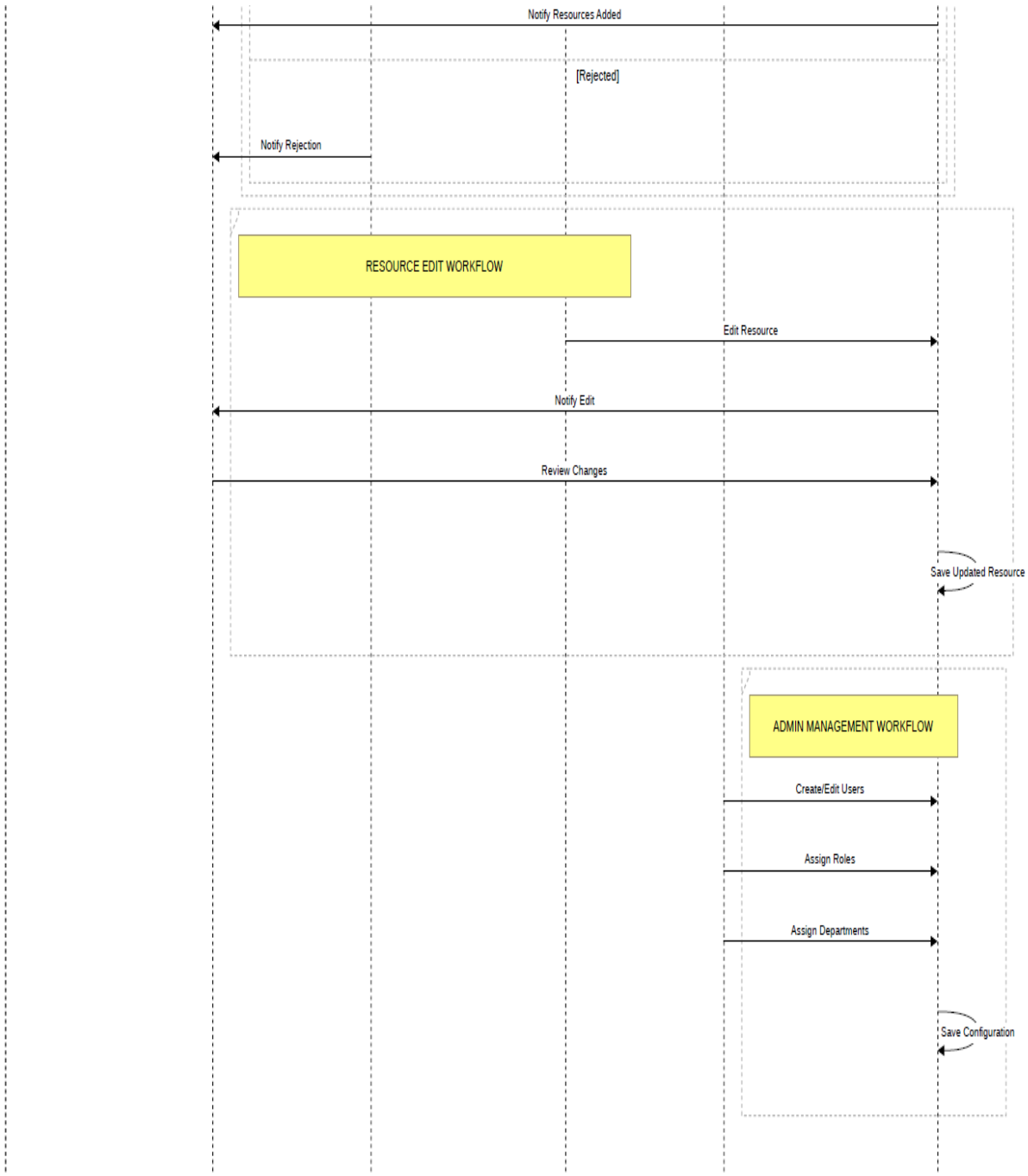
Sequence diagram



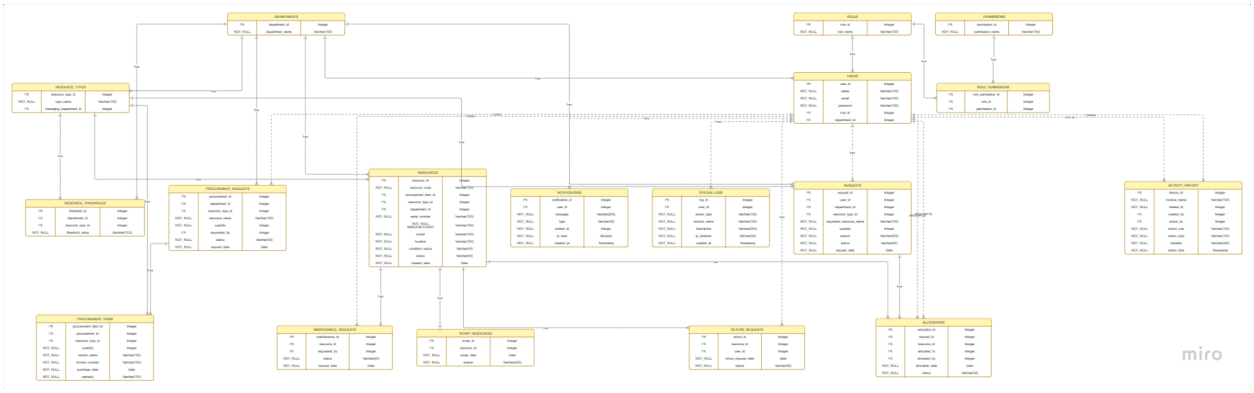

Add to My Resources
Move Request to History
[Rejected]
Send Rejection Notification
Move Request to History
MAINTENANCE WORKFLOW
Request Maintenance
Notify Maintenance Request
Inspect Resource
alt
[Repairable]
Mark Under Maintenance
Repair Resource
Notify Repair Completed
Confirm Allocation


Confirm Allocation
Status -> Allocated
[Not Repairable]
Mark Resource Scrap
Notify Scrap
Move to Scrap Records
RESOURCE RETURN WORKFLOW
Return Resource
Notify Return Request
Inspect Resource Condition
alt
[Good Condition]
Mark Available
Update Resource History
Confirm Return
[Damaged]


[Damaged]
Mark Scrap
Move to Scrap Records
Notify Scrap
LOW STOCK ALERT WORKFLOW
Notify Low Stock
Create Procurement Request
Send Procurement Request
PROCUREMENT WORKFLOW
Review Procurement Request
alt
[Approved]
Create Procurement Task
Purchase Resources
Enter Vendor & Invoice
Register Resource Codes
Notify Resources Added


Notify Resources Added
[Rejected]
Notify Rejection
RESOURCE EDIT WORKFLOW
Edit Resource
Notify Edit
Review Changes
Save Updated Resource
ADMIN MANAGEMENT WORKFLOW
Create/Edit Users
Assign Roles
Assign Departments
Save Configuration

# Entity Relationship (ER) Diagrams



### Platform-Level Data Entities (added in version 2.0)

The ER diagram above covers the organization workflow. Version 2.0 adds the following entities.

| Entity | Main attributes | Relationships |
|---|---|---|
| Organization | id, name, adminName, adminEmail, status (Pending, Active, Suspended), subscriptionPlanId, subscriptionExpiryDate, revenueGenerated, organization and admin proof documents | Has many users, departments and resources; has one plan; assigned to one Platform Employee |
| Subscription Plan | id, name (Starter, Professional, Enterprise), maxUsers, pricePerYear, features | Used by many organizations |
| Invoice | id, organizationId, planName, amount, date, status (Paid, Unpaid) | Belongs to one organization |
| Support Ticket | id, organizationId, title, description, status (Open, In Progress, Resolved), createdBy, date, reply | Raised by a System Admin; resolved by a Platform Employee |
| Announcement | id, targetOrgId (one organization or ALL), title, message, type (Maintenance, New Feature, Policy, General), authorId, date, adminReply | Sent by a Platform Employee or Owner to one or all organizations |
| Notification | id, organizationId, recipientRole, recipientUserId, department, title, description, read, time | Delivered to users by workflow events |
| User (updated) | role now one of the nine actors; department empty for Requestor, Registrar, Owner and Employee; organizationId empty for Owner and Employee; unique role-coded email | At most one Department Head per department |
| Resource (updated) | returnDate, extensionStatus (None, Pending, Approved, Rejected), extensionProposedReturnDate, extensionReason, extensionRequestedBy | Extension approved by the Department Head |

## 7. System Evolution

The system is designed to support future enhancements such as persistent database storage, online subscription payments, mobile access, advanced analytics, and integration with external systems.

# 8. Appendices

## 8.1 Hardware Requirements

- Server with minimum 8 GB RAM
- Client systems with modern web browsers

## 8.2 Software Requirements

- Web browser (Chrome, Firefox, Edge)
- Node.js 18 or higher with npm
- Backend: NestJS 11, TypeScript, Passport JWT, Swagger
- Frontend: React 19, Vite, React Router

# 9. Index


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- System Models – Section 6